

Decision Making Case study

Case 1 | Final-Year Project: Which Idea?

CASE DESCRIPTION

A team of four mechanical engineering students must choose one project:

- (A) solar-powered water pump,**
- (B) automated drone delivery, or**
- (C) low-cost prosthetic hand.**

They have six months, a limited budget and basic laboratory facilities.

The team initially prefers the drone because it sounds innovative.

QUESTIONS

1. Which criteria should be used to compare the three alternatives?
2. Create a simple weighted decision matrix using feasibility, time/resources, learning value, career alignment and impact.
3. **Does the most innovative option necessarily become the best decision? Why?**
4. Which option would you recommend and what assumption could change your decision?

Hint: Rational Model + Decision Matrix

Case 2 | CNC Breakdown: What Should We Do Now

CASE DESCRIPTION

The only CNC machine in the department breaks down two days before an important practical. The team can

- (A) pay for immediate repair,
- (B) borrow another department's machine, or
- (C) redesign the practical using another machine.

The budget is limited and the deadline cannot easily move.

QUESTIONS

1. Draw a decision tree showing the three options and their likely consequences.
2. Which branch gives the best balance of urgency, cost and learning continuity?
3. If only 30 minutes are available to decide, would your process change?
4. When would a 'good enough' decision be preferable to a fully analyzed decision?

KEY TAKEAWAY • The best decision process depends on both the problem and the time/information available.

Case 3 | Internship: Money Now or Skills Later?

CASE DESCRIPTION

A student receives two offers. Offer A is a well-paid IT internship with high stability but little connection to the student's core engineering interests. Offer B is a low-paid robotics internship with greater technical learning, networking and alignment with a desired career.

QUESTIONS

1. What criteria should matter most to this student?
2. Build a two-option decision matrix using financial support, skill relevance, career growth, networking and stability.
3. How would the decision change if the student urgently needed income?
4. Is there a single 'correct' decision? Explain.

Here, there is no single correct choice. The appropriate choice depends on the student's priorities, financial situation and career goals.

Case 4 | Job, M.Tech or Competitive Exam?

CASE DESCRIPTION

A final-year student has three paths:

- (A) accept an MNC job,
- (B) pursue M.Tech with a scholarship, or
- (C) prepare for UPSC/GATE.

The job offers immediate stability; MTech offers specialization; exam preparation offers potentially high long-term returns but substantial uncertainty.

QUESTIONS

1. Identify the student's short-term and long-term criteria.
2. Use a decision matrix for financial stability, career growth, risk, flexibility and personal aspiration.
3. Which decision would change if the student had major financial responsibilities?

Case 5 | Bridge Material: Rational Model in Action

CASE DESCRIPTION

Civil engineering students must choose material for a model footbridge:

- steel is strong but costly;
- aluminum is light with medium strength;
- composite material is innovative but needs more testing.

They need a defensible decision for their project review.

QUESTIONS

1. Step 1: Define the decision criteria. What should be included?
2. Step 2: Decide whether all criteria should have equal weight.
3. Step 3: Score each material and calculate a total.
4. Step 4: What would make you revisit the decision?

Hint

The team should consider strength/durability, cost, weight, availability and construction time. Weighted scoring is preferable when criteria have different importance. **The final choice should be supported by data rather than preference, and revised if key assumptions change.**

Case 6 | Tech Fest: The 'Good Enough' Motor Driver

CASE DESCRIPTION

Minutes before a robotics final, the team's motor driver fails. Available options are:

- (A) a lower-power driver that may overheat,
- (B) a slightly incompatible driver requiring rapid re-coding, or
- (C) a spare from another team. There is no time for complete testing.

QUESTIONS

1. What information is essential before deciding—and what can be ignored?
2. Define the minimum acceptable solution for the robot.
3. Which option is 'good enough' while remaining acceptably safe?
4. How is this different from using a full decision matrix?

Hint:

The team should identify non-negotiables: basic functionality, electrical safety and enough reliability to complete the round. **A slightly less optimal but workable option may be selected.** The key is recognising the constraints instead of pretending that a perfect analysis is possible.

Bounded Rationality + Satisficing

Case 7 | Circuit Failure: Intuition?

CASE DESCRIPTION

During an electronics demonstration, a circuit suddenly stops working.

A student without much lab experience immediately suspects the power supply issue, loose wiring or a burnt resistor. The student checks the power supply first and finds the fault. Afterwards, he/she says, working on intuition is the best option.

QUESTIONS

1. To what extent you agree with what the student says?
2. When can intuition become dangerous or misleading?
3. What quick verification step should follow an intuitive decision?

Case 8 | Electrical Fire: Expert Recognition Under Pressure

CASE DESCRIPTION

During a workshop, a small fire starts in a mechanical laboratory because of an electrical short circuit. An experienced lab assistant immediately recognizes the **pattern** and uses a CO₂ extinguisher rather than water. There is no time for a lengthy comparison of alternatives.

QUESTIONS

1. What pattern is the experienced person recognising?
2. Why is a formal decision matrix inappropriate in this situation?
3. What previous training makes the rapid decision possible?
4. What could happen if the pattern is misidentified?